

Bernie Calderon Magtalas

Infrastructure economist and licensed electrical engineer

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ABOUT

Infrastructure economist appraising energy, transport, and climate investments for the Philippine Government since 2016

34	15	48	8
infrastructure projects appraised, USD 14.05 billion	technical assistance and program financings reviewed	bills, laws, and policies reviewed	national boards and committees advised

Bernie Calderon Magtalas is an infrastructure economist and licensed electrical engineer. He is a Supervising Economic Development Specialist in the Infrastructure Staff of the Department of Economy, Planning, and Development (DEPDev) Investment Programming Group, where he leads project appraisal, planning, and policy formulation across the energy, transportation, and wider public infrastructure sectors.

He began his career at the Department of Energy (DOE) in December 2015, supporting renewable energy and rural electrification programs in remote, off-grid, and conflict-affected areas, and joined the National Economic and Development Authority (NEDA, now DEPDev) in September 2016.

Project economic analysis · Cost-benefit analysis · Infrastructure appraisal · Investment programming · Socioeconomic development planning · PPP risk assessment · Energy and transport policy · R · Python · Excel

SELECTED WORK

Highlights

ENERGY · PPP APPRAISAL LEAD

Agus-Pulangi Hydroelectric Power Plants Rehabilitation and Asset Management Project

USD 512 million

Leads the public-private partnership (PPP) appraisal of the Agus-Pulangi hydropower complex, a major source of clean power for the Mindanao grid.

CLIMATE FINANCE · POLICY-BASED LOAN

Climate Change Action Program, Subprogram 3

USD 650 million

Assessed the Asian Development Bank (ADB) policy-based loan, cofinanced by the Asian Infrastructure Investment Bank, Agence Française de Développement, and the Japan International Cooperation Agency, ahead of loan negotiations.

TRANSPORT · PPP RISK

PPP transport portfolio

USD 10 billion in contracts

Led the risk assessment of eight PPP transport projects and built mitigation into their contracts.

ANALYTICS · R

Regional Infrastructure Development Index

First in the Philippines

Directed the team of ADB consultants and DEPDev specialists that built the country's first composite infrastructure index.

CAREER

Experience



Department of Economy, Planning, and Development (DEPDev)

September 2016 to present

Formerly the National Economic and Development Authority (NEDA) · Infrastructure Staff, Pasig City

Appraises and evaluates energy, transport, and other infrastructure projects for Investment Coordination Committee (ICC) approval, including official development assistance (ODA) loans and public-private partnerships (PPPs), and reviews technical assistance proposals. Reviews bills and policies, represents DEPDev on inter-agency boards and technical working groups, and coordinates with multilateral development banks and development finance institutions.

- **Supervising Economic Development Specialist** — January 2025 to present
- **Senior Economic Development Specialist** — January 2020 to December 2024
- **Economic Development Specialist** — September 2016 to December 2019

See Appraisal portfolio · Policy and governance · Analytics · Engagements

Department of Energy (DOE)

December 2015 to August 2016

Taguig City

Supported renewable energy and rural electrification programs, including field evaluations of solar projects in remote areas with World Bank consultants.

- **Science Research Specialist**

See Field missions

APPRAISAL AND EVALUATION

Appraisal portfolio

Appraisal covered financial and economic models; market, technical, social, environmental, and institutional analyses; gender responsiveness; and, for public-private partnerships (PPP), review of concession agreements, risk allocation, and prequalification.

Projects

USD 14,047.33 million across 34 projects

PPP	LOCAL FUNDS	ODA
10,436.09	2,698.14	913.10
USD million · 74% · 16 projects	USD million · 19% · 13 projects	USD million · 7% · 5 projects

NO.	PROJECT	SUB-SECTOR	FINANCING	USD M
1	Unsolicited Build-Operate-Transfer (BOT) Proposal for the 50-Year Integrated Development Plan for the Mactan-Cebu International Airport	Airport	PPP	4,382.42
2	Quezon-Bicol Expressway Project	Roads and Bridges	Local funds	1,595.90
3	Unsolicited Build-Transfer-Operate and Maintain (BTOM) Proposal: Cebu Monorail Transit System Project	Rail	PPP	1,455.33
4	Unsolicited Operate-Add-Transfer (OAT) Proposal for the Laguindingan Airport	Airport	PPP	982.92
5	Unsolicited Operate-Add-Transfer (OAT) Proposal for the Davao International Airport	Airport	PPP	825.81
6	Metro Manila Intelligent Transport System (MMITS)	Urban	Local funds	610.07
7	Unsolicited Proposal for the Cavite-Tagaytay-Batangas Expressway (CTBEx) Project	Expressways	PPP	546.96
8	Agus-Pulangi Hydroelectric Power Plants Rehabilitation and Asset Management Project	Energy	PPP	512.00
9	Acquisition of Forty (40) Units of Fast Patrol Craft (FPC)	Safety	Local funds	471.69
10	Unsolicited Proposal for the Port of Davao (Sasa)	Ports	PPP	382.23

NO.	PROJECT	SUB-SECTOR	FINANCING	USD M
11	Unsolicited Operate-Add-Transfer (OAT) Proposal: New Bohol International Airport	Airport	PPP	361.05
12	New Dumaguete (Bacong) Airport	Airport	ODA	274.84
13	National Energy Efficiency and Conservation Program (NEECP)	Energy	ODA	260.00
14	Unsolicited Proposal for the Iloilo Commercial Port Complex (ICPC) and Port of Dumangas (POD)	Ports	PPP	225.62
15	Unsolicited Proposal for the Development, Operation and Management of the Bacolod-Silay International Airport	Airport	PPP	213.21
16	Agus-Pulangi Hydropower Plant Complex (APHC) Rehabilitation under Series of Project (SOP) 1	Energy	ODA	195.56
17	Philippine Geothermal Resource Derisking Facility (PGRDF)	Energy	ODA	175.00
18	Unsolicited Operate-Add-Transfer (OAT) Proposal for the Iloilo International Airport	Airport	PPP	126.14
19	Unsolicited Build-Own-Operate (BOO) Proposal for the North Luzon Express Terminal (NLET) Project	Urban	PPP	124.66
20	Unsolicited Proposal for the Puerto Princesa International Airport	Airport	PPP	107.61
21	Unsolicited Proposal for the Port of General Santos	Ports	PPP	101.54
22	Unsolicited Operate-Maintain-Upgrade-and-Transfer (OMUT) Proposal for the Kalibo International Airport	Airport	PPP	63.84
23	Unsolicited Proposal for the Busuanga Airport	Airport	PPP	24.75
24	Hybridization of Existing Diesel Power Plant in Zumarraga Island Mini-Grid with Solar PV, Wind Turbines, and Batteries	Energy	Local funds	7.74
25	Access to Sustainable Energy Programme (ASEP): Investment Support Component	Energy	ODA	7.70
26	Hybridization of Existing Diesel Power Plant in Biri Island Mini-Grid with Solar PV, Biomass, Wind Turbines, and Batteries	Energy	Local funds	4.51
27	Hybridization of Existing Diesel Power Plant in San Antonio Island Mini-Grid with Biomass, Solar PV, and Batteries	Energy	Local funds	1.79

NO.	PROJECT	SUB-SECTOR	FINANCING	USD M
28	Hybridization of Existing Diesel Power Plant in Capul Island Mini-Grid with Biomass and Batteries	Energy	Local funds	1.34
29	Hybridization of Existing Diesel Power Plant in Tagapul-an Island Mini-Grid with Biomass and Batteries	Energy	Local funds	1.12
30	Hybridization of Existing Diesel Power Plant in Santo Niño Island Mini-Grid with Biomass, Solar PV, and Batteries	Energy	Local funds	0.90
31	Hybridization of Existing Diesel Power Plant in Camandag Island Mini-Grid with Biomass and Batteries	Energy	Local funds	0.87
32	Hybridization of Existing Diesel Power Plant in San Vicente Island Mini-Grid with Biomass, Solar PV, and Batteries	Energy	Local funds	0.81
33	Hybridization of Existing Diesel Power Plant in Batag Island Mini-Grid with Biomass and Batteries	Energy	Local funds	0.73
34	Hybridization of Existing Diesel Power Plant in Almagro Island Mini-Grid with Biomass and Batteries	Energy	Local funds	0.67

34 of 34 projects shown Total USD 14,047.33 million

Technical assistance and programs

Reviews technical assistance (TA), grants, and policy-based loans for NEDA (now the Department of Economy, Planning, and Development, DEPDev) in response to development partners' requests for no-objection, comments, and endorsement.

USD 1,774.48 million across 15 programs

PROGRAMS AND POLICY-BASED LOANS
1,730.00
USD million · 97% · 3 programs

TECHNICAL ASSISTANCE AND GRANTS
44.48
USD million · 3% · 12 programs

Technical assistance and grants

NO.	PROGRAM	SECTOR	APPROVED	USD M
1	Advance Sustainable Clean Energy Network for Development (ASCEND) ADB TA 10312-REG	Energy	Jun 2024	4.65
2	Development of Renewable Energy Public-Private Partnership Projects and Transaction Advisory Services ADB TA 10375-REG	Energy	Nov 2024	3.00

NO.	PROGRAM	SECTOR	APPROVED	USD M
3	Accelerating Climate Action, Innovation, and Private Sector Development through Sector Operations ADB TA 10406-REG	Climate Change	Nov 2024	14.80
4	Advancing Energy Transition and Regional Power Trade in the Greater Mekong Subregion and Southeast Asia ADB TA 10480-REG	Energy	Dec 2024	5.00
5	Adaptation Metrics to Scale Up and Align Investment with Climate-Resilient Development ADB TA 10048-REG	Climate Change	Dec 2022	1.00
6	Scaling Up Clean Air Actions for Better Health and Resilience ADB TA 10537-REG	Environment	Apr 2025	4.13
7	Operations, Knowledge, and Analytics Support in Fragile and Conflict-Affected Situations and Small Island Developing States ADB TA 10045-REG	Governance	Dec 2022	3.75
8	Promoting Upstream Low-Carbon and Sustainable Environment Solutions in Southeast Asia ADB TA 10633-REG	Climate Change	Nov 2025	1.50
9	Catalyzing Growth and Emerging Areas in Asia and the Pacific ADB TA 0078-REG	Energy	Nov 2025	4.00
10	Readiness Support to Strengthen the Philippines' Capacity and Knowledge on Accessing the Green Climate Fund GCF PHL-RS-003	Climate Change	Sep 2019	0.30
11	Strengthening the Capacity of Institutions in the Philippines to Comply with the Transparency Requirements of the Paris Agreement (Capacity-building Initiative for Transparency, CBIT Philippines) GEF ID 11950	Climate Change	Jan 2026	1.95
12	Advancing Renewable Energy through Floating Solar Photovoltaics in the Philippines GGGI PH047	Energy	Started Jun 2025	0.40
12 programs				USD 44.48 million

Programs and policy-based loans

NO.	PROGRAM	SECTOR	APPROVED	USD M
1	**Climate Change Action Program (CCAP), Subprogram 3** ADB policy-based loan	Climate Change	Oct 2026 (DBCC)	650.00
2	**Philippines Second Energy Transition and Climate Resilience Development Policy Loan** World Bank P510184	Energy	Jun 2026	1,020.00
3	**Philippines Energy Threshold Program** Millennium Challenge Corporation (MCC)	Energy	Jun 2026	60.00

3 programs — USD 1,730.00 million

Costs are each program's total approved amount. Most technical assistance projects are regional, so the amount covers all participating countries, not only the Philippines.

Consulting studies

Evaluated the technical and financial proposals for all 12 studies, from procurement through contract negotiation. For two studies, also administered the consulting contract and evaluated the consultants' outputs.

USD 12.68 million across 12 studies · 2 administered through completion



MASTER PLANS	DEVELOPMENT PLANS	FEASIBILITY STUDIES
6.16	4.10	2.42
USD million · 49% · 6 studies	USD million · 32% · 3 studies	USD million · 19% · 3 studies

NO.	STUDY	LOCATION	USD M
1	**Feasibility Study (F/S) for the Pasig River Ferry System Project**	National Capital Region (NCR)	0.58
2	**F/S on Hybridization of Existing Diesel Power Plants (DPPs) with Renewable Energy (RE) and Battery** Contract administered	Region VIII	0.75
3	**Master Plan for the Sustainable Urban Infrastructure Development in Metropolitan Cagayan de Oro**	Region X	1.06
4	**Master Plan for Sustainable Urban Infrastructure Development in General Santos City**	General Santos City	1.12
	Total		**12.68**

NO.	STUDY	LOCATION	USD M
5	Master Plan for the Sustainable Urban Infrastructure Development in Vigan City Contract administered	Vigan City	0.81
6	Master Plan for the Sustainable Urban Infrastructure Development in Butuan City	Butuan City	1.06
7	Master Plan for the Sustainable Urban Infrastructure Development in Metropolitan Tuguegarao	Metropolitan Tuguegarao	1.08
8	National Broadband Plan (NBP) – Accelerated Fiber Build (Access) Project	Nationwide	1.09
9	Tourism Infrastructure Development Plans in Top Tourist Destinations in the Philippines Based on the Carrying Capacities	Nationwide	1.56
10	Tourism Infrastructure Development Plans for Transforming Communities Towards Resilient, Inclusive and Sustainable Tourism (TouRIST) Program	Siargao Island, Siquijor, Coron and El Nido, Palawan, Davao City and Samal Island, and Bohol	1.48
11	Integrated Infrastructure and Investment Corridors in Central Luzon	Region III	1.06
12	Master Plan for Sustainable Infrastructure Development in Olongapo and its Immediate Environs	Olongapo and its immediate environs	1.03
	Total		**12.68**

POLICY AND GOVERNANCE

Legislation, boards, and working groups

Bills, laws, and policies reviewed 48

MEASURE	DESCRIPTION	SECTOR
Amendments to the Electric Power Industry Reform Act (EPIRA): Strengthening the Energy Regulatory Commission 2025–2026 Senate and House measures, 20th Congress	Amends the Electric Power Industry Reform Act of 2001 (Republic Act No. 9136) to strengthen the regulatory powers and institutional capacity of the Energy Regulatory Commission.	Energy

MEASURE	DESCRIPTION	SECTOR
Waste-to-Energy Regulatory Framework Act 2025–2026 19th and 20th Congress versions	Sets a national energy policy and regulatory framework for facilities that use waste-to-energy technologies, implemented with local governments. A LEDAC priority measure.	Energy
Philippine National Nuclear Energy Safety Act 2025–2026 Senate Bill No. 2899 / House Bill No. 9293, 19th Congress; enacted as Republic Act No. 12305 (18 September 2025)	Provides a comprehensive legal framework for nuclear safety, security, and safeguards in the peaceful use of nuclear energy, and creates the Philippine Atomic Energy Regulatory Authority (PhilATOM).	Energy
Expansion of the Franchise Area of Davao Light and Power Company, Inc. 2025–2026 House Bill No. 11072, 19th Congress	Expands the franchise area of Davao Light and Power Company, Inc. to additional areas in the Davao Region.	Energy
Philippine Nuclear Liability Act 2025–2026 House Bill No. 375, 20th Congress	Sets the rules on civil liability and compensation for nuclear damage: who is liable, and how affected parties can claim compensation for incidents at nuclear installations or during the transport of nuclear materials.	Energy
Incentives for the Integration of the Nuclear Energy Sector into the Electricity Sector 2025–2026 House Bill No. 8567, 20th Congress	Prescribes incentives for integrating nuclear energy into the country's electricity sector and appropriates funds for this purpose.	Energy
Amendments to the Biofuels Act of 2006 (Republic Act No. 9367) 2025–2026 Pending House Bills, 20th Congress	Amends the Biofuels Act of 2006 (Republic Act No. 9367), which mandates the blending of biofuels into locally sold gasoline and diesel.	Energy

MEASURE	DESCRIPTION	SECTOR
Amendments to the Downstream Oil Industry Deregulation Act (Republic Act No. 8479) and a Strategic Fuel Reserve 2025–2026 Senate and House measures, 20th Congress	Amends the Downstream Oil Industry Deregulation Act of 1998 (Republic Act No. 8479) and establishes a strategic fuel reserve to manage supply and price disruptions.	Energy
Sariling Kuryente Act: Behind-the-Meter Generation, Energy Storage, and Rooftop Solar 2025–2026 Substitute bill, House Committee on Energy, 20th Congress	Promotes self-generation of electricity by households and businesses through behind-the-meter generation, energy storage systems, and rooftop solar.	Energy
Franchise Renewal of Visayan Electric Company, Inc. 2025–2026 Senate Committee on Public Services, 20th Congress	Renews the legislative franchise of Visayan Electric Company, Inc. to distribute electricity in its Metro Cebu service area.	Energy
Franchise Coverage Expansion of MORE Electric and Power Corporation 2025–2026 Senate Committee on Public Services, 20th Congress	Expands the franchise coverage of MORE Electric and Power Corporation, the electricity distributor of Iloilo City.	Energy
Draft Administrative Order: Accelerating the Transition to Renewable Energy through the Use of Electric Vehicles in Mass Transport 2025–2026 Office of the President; Department of Transportation	Accelerates the shift to renewable energy by promoting the use of electric vehicles in public mass transport.	Transportation

MEASURE	DESCRIPTION	SECTOR
IAEECC Resolution No. 13: Guidelines for Government Entities as Eligible End-Users under the Customer Choice Programs in the Retail Market 2025–2026 Inter-Agency Energy Efficiency and Conservation Committee; signed September 2026	Sets guidelines allowing government entities to choose their electricity supplier as eligible end-users under the customer choice programs of the retail electricity market.	Energy
IAEECC Resolution Directing Government Entities to Prioritize the Purchase or Rental of Electric Vehicles 2025–2026 Inter-Agency Energy Efficiency and Conservation Committee	Directs government entities to prioritize electric vehicles when buying or renting motor vehicles, in line with the Electric Vehicle Industry Development Act (Republic Act No. 11697).	Energy
Amendments to the Electric Vehicle Recognition Guidelines (Department Circular No. DC2023-05-0012) 2025–2026 Department of Energy	Amends the Department of Energy's guidelines for recognizing and classifying electric vehicles under the Electric Vehicle Industry Development Act.	Energy
Framework for the Integration of Nuclear Energy in the Clean Energy Scenario of the Philippine Energy Plan 2023–2050 2025–2026 Department of Energy, draft Department Circular	Sets the framework for integrating nuclear power into the clean energy scenario of the Philippine Energy Plan 2023–2050.	Energy
Nuclear Capacity Tender Guidelines 2025–2026 Department of Energy, draft Department Circular	Sets the guidelines for competitive tenders to procure nuclear generating capacity.	Energy
National Nuclear Security Policy and Strategy 2025–2026 NEP-IAC Subcommittee 2	Sets the national policy and strategy for securing nuclear and radioactive materials and facilities.	Energy

MEASURE	DESCRIPTION	SECTOR
Amendments to the Implementing Guidelines of the Philippine Transport Vehicles Fuel Economy Labeling Program 2025–2026 Department of Energy	Revises the registration procedures and compliance mechanisms of the program that requires fuel economy labels on vehicles.	Transportation
Amendment to DHSUD Department Circular No. 2024-004: Locational Clearance for Projects of National Significance 2025–2026 Department of Human Settlements and Urban Development	Amends the guidelines for processing locational clearances for projects declared of national significance.	Governance
Carbon Pricing Instruments for the Philippines 2025–2026 Department of Finance; Carbon Pricing Support TWG	Assesses legal and policy options for carbon pricing instruments in the Philippines.	Climate Change
Waste-to-Energy Act Senate Bill No. 363	An Act Establishing a National Energy Policy and Regulatory Framework for Facilities Utilizing Waste-to-Energy Technologies	Energy
Waste-to-Energy Act of 2019 Senate Bill No. 401	An Act Allowing the Use of Waste to Energy Technology in Electricity, Fuel and Heat Generation, and for other Purposes	Energy
Rural Airport Development Act House Bill No. 5075	An Act Granting the Right to Franchisees to Develop the Rural Airport Infrastructure Network in the Philippines for Commercial and for Other Purposes	Transportation
Organic Law for the Bangsamoro Autonomous Region in Muslim Mindanao Republic Act No. 11054	An Act Providing for the Organic Law for the Bangsamoro Autonomous Region in Muslim Mindanao, Repealing for the Purpose Republic Act No. 6734, Entitled “An Act Providing for an Organic Act for the Autonomous Region in Muslim Mindanao,” as Amended by Republic Act No. 9054, Entitled “An Act to Strengthen and Expand the Organic Act for the Autonomous Region in Muslim Mindanao”	Governance

MEASURE	DESCRIPTION	SECTOR
One Million Solar Roofs Act of 2017 House Bill No. 6565	An Act Providing for Incentives and Credit Facilities for Consumers of Small Solar Power Systems in the Philippines	Energy
Senate Bill No. 1007	An Act to Reconstruct and Rehabilitate the Existing 479-kilometer PNR lines from San Fernando, La Union to Legazpi City, Albay, Providing Funds from the Annual Value-Added Tax Collections, Amending for the Purpose the National Internal Revenue Code of 1997, As Amended	Transportation
Philippine Downstream Natural Gas Industry Development Act (enacted as RA No. 12120, the Philippine Natural Gas Industry Development Act, 8 January 2025) House Bill No. 10649	An Act Promoting the Development of the Philippine Downstream Natural Gas Industry, Consolidating for the Purpose All Laws Relating to the Transmission, Distribution and Supply of Natural Gas, and Appropriating Funds Therefor	Energy
Senate Resolution No. 359	Resolution Directing the Senate Committee on Energy to Conduct an Inquiry in Aid of Legislation on the Sufficiency of the Department of Energy's Existing Policy on the Minimum Inventory Requirement for Oil and Petroleum Products and the Necessity of Establishing a Strategic Petroleum Reserve to Ensure the Security of the Country's Oil Supply	Energy
Senate Resolution No. 360	A Resolution Directing the Senate Committee on Energy to Conduct an Inquiry on the Implementation of Republic Act No. 8479 Otherwise Known as the Downstream Oil Industry Deregulation Act of 1998 With the End in View of Introducing Remedial Legislation as May Be Necessary	Energy
Air Carriers Relief Act House Bill No. 9324	An Act Providing for Relief Package to the Air Transport Sector of the Philippines	Transportation
Electric Vehicles and Charging Stations Act Senate Bill No. 1382	An Act Providing the National Energy Policy and Regulatory Framework for the Use of Electric Vehicles and the Establishment of Electric Charging Stations	Energy

MEASURE	DESCRIPTION	SECTOR
Electric Vehicle Industry Development Act House Bill No. 10213	An Act Providing for the Development of the Electric Vehicle Industry and Establishing a Regulatory Framework for the Manufacture and Use of Electric Vehicles	Energy
Energy Efficiency and Conservation Act Republic Act No. 11285	An Act Institutionalizing Energy Efficiency and Conservation, Enhancing the Efficient Use of Energy, and Granting Incentives to Energy Efficiency and Conservation Projects	Energy
Philippine Energy Research and Policy Development Institute Act of 2016 Senate Bill No. 1290	An Act Establishing the Philippine Energy Research and Policy Development Institute, Defining Its Objectives, Powers and Functions, Providing Funds Therefor, and for Other Purposes	Energy
Tax Reform for Acceleration and Inclusion (TRAIN) Senate Bill No. 1592	An Act Amending Sections 5, 6, 24, 25, 31, 34, 35, 51, 79, 84, 86, 89, 90, 97, 99, 100, 101, 106, 107, 108, 109, 112, 114, 116, 148, 149, 150, 151, 155, 171, 196, 232, 237, 249, 264, and 288; Creating New Sections 148-A, 150-A, 237-A, 264-A, 264-B, and 265-A; All Under Republic Act No. 8424, Otherwise Known as the National Internal Revenue Code of 1997, as Amended, and for Other Purposes	Trade and Industry
The Solar Energy in National Government Offices Act of 2016 House Bill No. 2172	An Act Reducing the Cost of Electricity for National Government Offices by Promoting the Development and Utilization of Solar Energy in All Government Building and Offices, Providing Funds Therefor, and for Other Purposes	Energy
Solar Energy Act of 2016 House Bill No. 5402	An Act Mandating Government Offices and/or Establishments to Allot at Least Thirty Percent (30%) of their Energy Requirements to Solar Energy, and Corollarily, Mandating the Same to Install Solar Panels unto Their Buildings	Energy
Small Solar Power System Incentives Act House Bill No. 6960	An Act Promoting the Use of Renewable Energy in Homes by Providing Incentives and Credit Facilities for Consumers Utilizing Small Solar Power Systems	Energy
House Bill No. 8723	An Act Strengthening the Protection and Management of Waterways and Water Systems by Host Communities, and for Other Purposes	Water

MEASURE	DESCRIPTION	SECTOR
House Bill No. 1484	An Act Requiring All Independent Power Producers, Generation Companies or Energy Resource Developers to Remit the Amount They Are Required to Set Aside as Financial Benefit Directly to the Host Community	Energy
House Bill No. 1388	An Act Providing for the Development, Rehabilitation and Modernization of Naval Airport in the Province of Biliran and to Allocate Funds Thereof	Transportation
Ubay Airport Development Act House Bill No. 2112	An Act Providing for the Ubay Airport Development to Accommodate Commercial Domestic and Aviation Training Flights, Appropriating Funds Therefor, and for Other Purposes	Transportation
House Bill No. 2769	An Act Providing for the Establishment and Construction of an International Airport in the Center of Mindanao, Particularly in the Are Between Damulog, Bukidnon in the North; Antipas, North Cotabato in the East; Carmen, North Cotabato in the South; And Alamada, North Cotabato in the West, and Appropriating Funds Therefor	Transportation
House Bill No. 4238	An Act Providing for the Establishment and Construction of an Airport in the Municipality of Luuk, Province of Sulu, for the Benefit of the Entire Province of Sulu, to Be Known as Luuk Airport, and Appropriating Funds Therefor	Transportation
House Bill No. 4393	An Act Designating Mati Airport in Davao Oriental as the Official Alternate Airport to the Davao International Airport and Appropriating Funds Therefor	Transportation
House Bill No. 7847	An Act Establishing a Domestic, Commercial Airport in the Municipality of San Francisco, Province of Cebu, and Appropriating Funds Therefor	Transportation
The Maria Cristina Falls Act of the Philippines House Bill No. 2317	An Act Guaranteeing the Right to the Natural Wealth and Granting a Twenty Percent Discount (20%) in the Electricity Rates to All Electric Consumers in Iligan City, Lanao Del Norte, And Lanao Del Sur being the Host City and Provinces of Maria Cristina Falls, Agus River, and Lake Lanao	Energy

Boards, committees, and working groups

Energy policy and security

Boards and committees · 5 bodies

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Energy Investment Coordinating Council	EICC
Inter-Agency Energy Contingency Committee	IECC
Nuclear Energy Program Inter-Agency Committee	NEP-IAC
Inter-Agency Energy Contingency Committee-Oil Contingency Task Force	IECC-OCTF
Inter-Agency Energy Efficiency and Conservation Committee	IAEECC

Power sector and infrastructure

Boards and committees · 3 bodies

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National Power Corporation Board of Directors	NPC
Infrastructure Development Committee	InfraCom
Power Sector Assets and Liabilities Management Corporation Board	PSALM

Sector policy

Working groups · 4 bodies

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Electric Vehicle Industry Development Act TWG	EVIDA
Sub-Committee on Sustainable Development Goals TWG on Environment	SC-SDGs
Technical Panel for Evaluation of Innovative Energy Projects	
Philippines' COVID-19 Recovery Cluster	

Within DEPDev

Working groups · 3 bodies

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Climate Change Community of Practice	COP
DEPDev Bids and Awards Committee for Goods and Infrastructure Projects TWG	BAC-GIP
Gender and Development Focal Point System	GAD

DATA ANALYTICS

Analytical tools

These tools were built in R, Python, and Excel to support infrastructure appraisal, investment programming, and energy policy research.

Regional Infrastructure Development Index (RIDI)

Open RIDI ↗

Directed five ADB consultants and five DEPDev specialists across the energy, water, digital, transport, and social sectors, and computed and analyzed the Philippines' first composite measure of infrastructure development in R and Python. Its interactive dashboard was adopted as a prototype for regional and sectoral assessments of infrastructure investment gaps.

Infrastructure Indicator Screening Dashboard

Open dashboard ↗

The dashboard was used to screen and select the indicators included in the Regional Infrastructure Development Index.

Multi-Criteria Socioeconomic Evaluation Model

207 projects

This automated Excel model prioritized 207 infrastructure projects for a USD 176 billion national investment pipeline.

ENGAGEMENTS

Knowledge networks and forums



Panelist · 26 February 2026, Bangkok

SIPA Regional Knowledge Network: session at the 13th Asia-Pacific Forum on Sustainable Development

The session, *Sustainable Infrastructure, Resilient Links: Energy, Transport and Industry in Asia's Low-Carbon Future*, was part of a series of knowledge-sharing activities on sustainable infrastructure run by the Asian Institute of Technology, the United Nations Development Programme, and the OECD under the Sustainable Infrastructure Programme in Asia (SIPA), funded by Germany's International Climate Initiative. The session stressed embedding sustainability in infrastructure planning and early-stage appraisal. OECD Environment post on the event ↗



Resource person and participant · Asian Institute of Technology, Thailand · 2025–2026

SIPA Regional Knowledge Network: Southeast Asia Training Seminars on Sustainable Infrastructure Development

SIPA is a regional network of knowledge institutions on sustainable infrastructure, connecting policymakers, practitioners, and experts in Indonesia, the Philippines, and Thailand. It is run by the Asian Institute of Technology, the United Nations Development Programme, and the OECD under the Sustainable Infrastructure Programme in Asia (SIPA).

First SIPA Southeast Asia Training Seminar on Sustainable Infrastructure Development
In person, AIT · 28 April–2 May 2025 ·

Second SIPA Southeast Asia Training Seminar on Sustainable Infrastructure Development
In person, AIT · 28 July–1 August 2025 ·

Third SIPA Southeast Asia Training Seminar on Sustainable Infrastructure Development
In person, AIT · 22–26 February 2026 ·

SIPA webinar series (9 sessions, online)
June 2025–March 2026 · financial and economic evaluation, risk and uncertainty, ecosystem services, pricing, economy-wide impacts, governance, and integrated frameworks



Panel discussant · 2024

Ease of Doing Business Forum 2024

Executive Order No. 59: Streamlining the Permitting Process for Infrastructure Flagship Projects



Future Energy Leader (FEL-100) · 2023–2024

World Energy Council

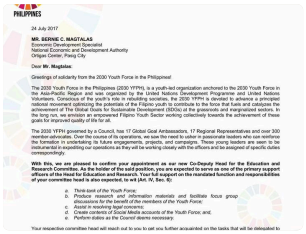
Nominated with the support of the Japan Energy Association and selected as a Future Energy Leader at the 26th World Energy Congress, Rotterdam, 22–25 April 2024. Contributed regional commentary to the World Energy Trilemma 2024 report.



Panel discussant

Socioeconomic Development Planning Advocacy

Took part in a briefing on infrastructure development principles, policies, and processes with the Asian Development Bank and the University of Tokyo.



Co-Deputy Head for Education and Research

2030 Youth Force in the Philippines Inc.

The organization works toward the Sustainable Development Goals at the grassroots.

FIELD WORK

Field missions

2016 · Took part in sustainability assessments of off-grid solar home system (SHS) programs with DOE, the National Electrification Administration (NEA), and the World Bank, ahead of the scaled-up Access to Sustainable Energy Programme (ASEP).

8–12 July 2016 · Negros Island Region

Photovoltaic Mainstreaming projects, three electric cooperatives

19–21 July 2016 · Infanta, Pangasinan

HEART and SOUL Project: 267 solar home systems in two barangays

Assessed in Barangay Pita with DOE, NEA, the World Bank, and Innovation Energie Développement (IED).

EXPERTISE

Methods and tools

Appraisal methods

These methods are applied under the NEDA (now DEPDev) Reference Manual on Project Development and Evaluation and the Investment Coordination Committee (ICC) guidelines.

APPRAISAL FRAMEWORK	Integrated project analysis across market, technical, financial, economic, institutional, social, and risk modules; incremental with-and-without-project comparison; least-cost alternatives across technology, location, scale, and timing

ECONOMIC ANALYSIS	Shadow pricing with ICC parameters (10% social discount rate, shadow exchange and wage rates); conversion factors, removal of transfers, valuation of externalities; EIRR, ENPV, and benefit-cost ratio
FINANCIAL ANALYSIS	Real and nominal cash-flow models from the owner, total-investment, and government budget viewpoints; FIRR against the weighted average cost of capital; annual liquidity and debt-service checks
RISK AND SENSITIVITY	ICC sensitivity cases (costs +10% and +20%, benefits −10% and −20%, combined), switching values; PPP risk identification, allocation, and mitigation costing
SOCIAL AND DISTRIBUTIONAL	Stakeholder analysis reconciling financial and economic results; social impacts and willingness to pay
CLIMATE AND ENVIRONMENT	Greenhouse gas abatement costing, climate resilience, nature-related impacts

Planning, policy, and research

INVESTMENT PROGRAMMING	Multi-criteria analysis to prioritize projects in the national infrastructure investment pipeline
POLICY ANALYSIS	Comparative analysis of legislation and regulatory frameworks, such as benchmarking the natural gas industry bill against the electric power industry's market framework
STATISTICS	Confirmatory factor analysis, structural equation modeling

Tools and languages

ANALYSIS	R, Python, Excel (economic and financial models)
VISUALIZATION	Interactive web dashboards built from R and Python outputs
ENERGY MODELING	OSeMOSYS, FlexTool (Climate Compatible Growth training, 2025)
LANGUAGES	Filipino (native), English (fluent)
LOCATION	Pasig City, Philippines

ACADEMIC BACKGROUND

Education

2024

Master of Arts in International Relations

Waseda University, Graduate School of Asia-Pacific Studies · Tokyo, Japan

September 2022 to September 2024 ·
Specialization in Economic Development and Policy in Asia

SCHOLARSHIP — Fully funded JICA Sustainable Development Goals (SDGs) Global Leader Program

THESIS — **Charting the Path to Net Zero ↗**

Built confirmatory factor analysis and structural equation models in R across 27 variables and 120 countries to measure how political, economic, and environmental factors shape the global energy transition.

COURSEWORK — Analysis of Quantitative Data I and II · Multivariate Statistics · Data Science for Management · Environmental Economics and Environmental Policy · Sustainable Development and International Development Cooperation

2014

Bachelor of Science in Electrical Engineering

University of the Philippines Los Baños, College of Engineering and Agro-Industrial Technology · Laguna, Philippines

May 2008 to August 2014 · Major in Power Engineering

CREDENTIALS

License and certifications

LICENSE

Registered Electrical Engineer

Professional Regulation Commission, Philippines · April 2015

CERTIFICATION

Certified PPP Professional (CP3P), Foundation

APMG International · January 2021

CERTIFICATION

Certified Flexible Power System Specialist (CFPSS)

Renewables Academy (RENAC) · November 2018

CERTIFICATION

Certified Green Finance Specialist

Renewables Academy (RENAC) · November 2017

COURSE

Energy and Flexibility Modelling: OSeMOSYS and FlexTool

Climate Compatible Growth · June 2025

COURSE

Modelling, Policy and Political Economy

Climate Compatible Growth · May 2025

REFERENCE

Acronyms and abbreviations

ADB	Asian Development Bank
AFD	Agence Française de Développement
AIIB	Asian Infrastructure Investment Bank
AIT	Asian Institute of Technology
ASEP	Access to Sustainable Energy Programme
BAC-GIP	Bids and Awards Committee for Goods and Infrastructure Projects
BOO	build-own-operate
BOT	build-operate-transfer
BTOM	build-transfer-operate and maintain
CAAP	Civil Aviation Authority of the Philippines
CALAx	Cavite-Laguna Expressway
CBIT	Capacity-building Initiative for Transparency
CCAP	Climate Change Action Program
CFL	compact fluorescent lamp
CFPSS	Certified Flexible Power System Specialist
COP	Community of Practice
CP3P	Certified Public-Private Partnership Professional
CTBEx	Cavite-Tagaytay-Batangas Expressway
DBCC	Development Budget Coordination Committee
DC	Department Circular
DED	detailed engineering design
DEPDev	Department of Economy, Planning, and Development
DHSUD	Department of Human Settlements and Urban Development
DOE	Department of Energy
DOTr	Department of Transportation
DPP	diesel power plant
EC	electric cooperative
EICC	Energy Investment Coordinating Council
EIRR	economic internal rate of return
ENPV	economic net present value
EO	Executive Order
EPIRA	Electric Power Industry Reform Act of 2001
ERC	Energy Regulatory Commission
EU	European Union
EVIDA	Electric Vehicle Industry Development Act
F/S	feasibility study
FEL-100	Future Energy Leaders 100 (World Energy Council)
FIRR	financial internal rate of return
FPC	fast patrol craft
GAD	Gender and Development
GCF	Green Climate Fund
GEF	Global Environment Facility
GGGI	Global Green Growth Institute
GIS	geographic information system
GOP	Government of the Philippines
HPP	hydroelectric power plant
IAEECC	Inter-Agency Energy Efficiency and Conservation Committee
IBRD	International Bank for Reconstruction and Development (World Bank)
ICC	Investment Coordination Committee
IECC	Inter-Agency Energy Contingency Committee
IECC-OCTF	Inter-Agency Energy Contingency Committee – Oil Contingency Task Force
IED	Innovation Energie Développement
ILS	Integrated Logistics Support
InfraCom	Infrastructure Development Committee
ITS	intelligent transport system
JICA	Japan International Cooperation Agency
KEXIM	Export-Import Bank of Korea
LANDBANK	Land Bank of the Philippines
LED	light-emitting diode
LEDAC	Legislative-Executive Development Advisory Council
MCC	Millennium Challenge Corporation
MMITS	Metro Manila Intelligent Transport System
MPFRB	Multi-Purpose Fast Response Boat
MVIP	Mindanao-Visayas Interconnection Project
MWp	megawatt-peak
NEA	National Electrification Administration
NEDA	National Economic and Development Authority (now DEPDev)
NEECP	National Energy Efficiency and Conservation Program
NEP-IAC	Nuclear Energy Program Inter-Agency Committee
NIR	Negros Island Region
NLET	North Luzon Express Terminal
NOCECO	Negros Occidental Electric Cooperative
NONECO	Northern Negros Electric Cooperative
NORECO I	Negros Oriental I Electric Cooperative
NPC	National Power Corporation
O&M	operation and maintenance
OAT	operate-add-transfer
ODA	official development assistance
OECD	Organisation for Economic Co-operation and Development
OMUT	operate-maintain-upgrade-and-transfer
PAP	programs, activities, and projects
PCG	Philippine Coast Guard
PGRDF	Philippine Geothermal Resource Derisking Facility
PhilATOM	Philippine Atomic Energy Regulatory Authority
PLLO	Presidential Legislative Liaison Office
PNR	Philippine National Railways

PPP	public-private partnership
PRC	Professional Regulation Commission
PSALM	Power Sector Assets and Liabilities Management Corporation
PTB	passenger terminal building
PV	photovoltaic
PVM	Photovoltaic Mainstreaming
RA	Republic Act
RENAC	Renewables Academy
RFID	radio-frequency identification
RIDI	Regional Infrastructure Development Index
RORO	roll-on/roll-off
SC-SDGs	Sub-Committee on the Sustainable Development Goals
SDGs	Sustainable Development Goals
SHS	solar home system
SIPA	Sustainable Infrastructure Programme in Asia
SOP	Series of Projects
TA	technical assistance
TASF	Technical Assistance Special Fund
TEFI	Team Energy Foundation, Inc.
TEU	twenty-foot equivalent unit
TWG	technical working group
UNDP	United Nations Development Programme
WTE	waste-to-energy